

COLLINGWOOD, ON, Canada

WMO: 712700

Lat: 44.505N

Lon: 80.223W

Elev: 180

StdP: 99.18

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-16.1	-22.0	0.5	-18.2	-19.3	0.7	-15.4	12.9	-7.4	11.8	-8.6	2.9	150	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	30.1	22.1	28.3	21.1	26.7	20.4	23.2	27.9	22.3	26.6	21.5	25.4	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.7	16.7	25.8	20.9	15.9	24.9	20.0	15.0	24.3	69.8	28.0	66.5	26.7	63.4	25.4	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.2	7.9	DB	-22.7	33.4	3.6	1.5	-25.3	34.4	-27.4	35.3	-29.5	36.1	-32.1	37.1	(4)
(5)			WB	-23.0	25.0	3.6	1.1	-25.5	25.8	-27.6	26.4	-29.6	27.0	-32.2	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-5.2	-4.9	-0.7	5.8	12.0	17.5	20.7	20.3	16.7	10.5	4.0	-1.7	(6)	
	DBStd	10.39	6.26	5.75	5.67	5.12	4.73	3.93	3.15	2.97	4.04	4.61	4.82	4.88	(7)	
	HDD10.0	2001	471	417	334	147	30	1	0	0	1	50	187	363	(8)	
	HDD18.3	4014	728	650	589	377	205	63	12	14	78	248	429	621	(9)	
	CDD10.0	1266	0	0	3	22	93	224	332	318	201	64	8	1	(10)	
	CDD18.3	239	0	0	0	1	10	37	86	73	28	4	0	0	(11)	
	CDH23.3	1516	0	0	1	13	83	261	529	426	182	21	0	0	(12)	
	CDH26.7	361	0	0	0	2	22	68	122	105	41	3	0	0	(13)	
(14) Wind	WSAvg	3.4	4.0	3.8	3.9	3.9	3.2	2.7	2.7	2.7	2.9	3.4	3.8	3.9	(14)	
(15) Precipitation	PrecAvg	956	90	68	65	69	80	80	83	74	86	89	94	87	(15)	
	PrecMax	1118	146	130	165	128	168	160	136	127	138	161	164	151	(16)	
	PrecMin	754	51	24	24	31	28	24	31	39	21	28	37	36	(17)	
	PrecStd	105	25	26	30	28	32	32	29	24	26	36	31	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	12.0	18.6	25.4	29.4	31.6	32.2	32.4	30.5	25.8	18.8	12.9	(19)	
		MCWB	10.1	8.5	13.4	16.6	20.0	22.4	22.9	23.2	22.3	17.9	13.5	10.8	(20)	
	2%	DB	7.8	8.1	13.7	20.8	25.6	28.6	29.6	29.3	27.5	22.9	15.9	8.9	(21)	
		MCWB	5.9	5.5	10.3	14.1	18.2	21.3	22.2	21.7	20.4	17.6	12.5	6.9	(22)	
	5%	DB	4.9	4.6	10.1	17.3	22.5	26.2	27.8	27.3	25.3	20.3	13.5	6.5	(23)	
		MCWB	3.4	2.4	7.1	12.0	16.5	19.9	21.2	21.0	19.6	15.8	10.4	4.7	(24)	
	10%	DB	2.8	2.5	6.7	13.9	19.7	24.2	26.3	25.5	23.4	17.7	11.1	4.4	(25)	
		MCWB	1.6	0.8	4.0	9.3	14.9	19.0	20.7	20.2	18.7	14.0	8.7	2.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	9.0	14.3	18.0	21.3	23.4	24.6	24.6	23.2	20.8	14.9	11.0	(27)	
		MCDB	11.6	10.6	17.8	22.7	27.5	28.9	30.0	29.4	28.3	24.4	16.9	13.1	(28)	
	2%	WB	6.1	5.7	10.7	14.9	19.4	22.2	23.3	22.9	21.8	18.2	12.7	7.3	(29)	
		MCDB	7.4	8.3	13.3	20.2	24.3	27.3	27.7	27.2	25.5	21.6	15.2	8.7	(30)	
	5%	WB	3.5	2.6	7.4	12.4	17.5	21.0	22.4	21.9	20.5	16.2	10.5	4.8	(31)	
		MCDB	4.7	4.3	10.3	16.8	21.8	25.0	26.4	25.9	24.2	19.7	13.2	6.3	(32)	
	10%	WB	1.6	0.9	4.3	9.7	15.4	19.6	21.4	21.0	19.2	14.5	8.8	2.9	(33)	
		MCDB	2.6	2.3	6.4	13.4	19.0	23.3	25.2	24.6	22.6	17.4	11.1	4.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.2	7.4	8.5	9.1	8.7	8.3	8.3	8.8	7.5	6.4	5.7	(35)	
		MCDBR	8.4	9.6	11.8	14.9	13.2	12.1	10.3	11.1	11.4	11.4	10.3	7.9	(36)	
	5% WB	MCWBR	7.4	8.0	8.9	9.6	7.8	6.6	5.6	5.7	6.2	7.0	7.9	6.7	(37)	
		MCWBR	8.2	9.6	11.6	13.9	12.5	10.7	8.9	9.1	9.8	10.1	9.6	7.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.284	0.302	0.317	0.358	0.388	0.410	0.407	0.402	0.370	0.347	0.314	0.287	(40)	
		taud	2.361	2.286	2.396	2.347	2.323	2.282	2.317	2.322	2.405	2.442	2.480	2.416	(41)	
	Ebn at Noon		836	883	921	904	882	860	858	850	852	824	800	798	(42)	
		Edh at Noon	77	99	102	118	124	130	124	119	102	86	69	66	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.26	2.22	3.52	4.51	5.41	5.92	5.98	5.10	3.99	2.31	1.33	0.89	(44)	
	RadStd		0.15	0.29	0.37	0.43	0.52	0.42	0.35	0.27	0.37	0.26	0.20	0.12	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page